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22MA201/ COMMON TO ALL BRANCHES (EXCEPT CSBS)

UNIT I – FIRST ORDER LINEAR DIFFERENTIAL EQUATIONS

LP 06 – EXPONENTIAL GROWTH

UNIT 1

MODELING POPULATION DYNAMICS OF SINGLE SPECIES

A mathematical model in population dynamics represents the pattern of growth of a given population in the presence of various environmental factors. The mathematical models in population dynamics can be classified based on various factors like nature of growth rate, its fluctuation, environmental factors, size of the population and so on.

We shall now discuss in brief some of the concepts which we shall be using for the study in this unit.

The dynamics of single population are generally described in terms of one-dimensional differential equations. Here we consider one – dimensional population models that were developed over the past several centuries to describe the growth and / or decay of single homogeneous populations.

As we have discussed, differential equations are commonly used in science to model the behavior of dynamical systems. In this chapter, we consider some of the simplest possible behaviors for a dynamical system: growth , decay and oscillation.

Of course, the simplest model for growth is linear growth, for which a variable 'y' that increases at a constant rate. This corresponds to the differential equation $\frac{dy}{dt} = r$ Where 'r' is the (constant) growth rate. The solutions to the equation are linear functions $y = y_0 + rt$

Note that the growth rate 'r' is the slope of the linear function, and the constant term $y_0 = y(0)$ is the initial value of 'y'. We are quite familiar with linear growth, and we will not discuss it further.

The second simplest model for growth is exponential growth, which corresponds to the differential equation

$$\frac{dy}{dt} = ky$$

The general solution of this differential equation is given below.

Proof:

$$y' = ky$$
$$\frac{y'}{y} = k$$
$$\int \frac{y'}{y} dt = \int k dt$$
$$\int \frac{1}{y} dy = \int k dt$$
$$\ln y = kt + C_1$$
$$y = e^{kt} e^{C_1}$$
$$y = Ce^{kt}$$
$$y(t) = Ce^{kt}$$

Where C can be determined if y(t) is given at certain time.

The values of *C* and *k* are determined with the help of initial conditions. Thus, the population increases exponentially with time. The law of population growth is called **Malthusian law**.

Exponential Growth and Decay Model

If y is a differentiable function of 't' such that y > 0 and $y' = ky$ for some constant 'k' , then $y = Ce^{kt}$ where 'C' is the initial value of 'y', and 'k' is the proportionality constant,

Exponential growth occurs when $k > 0$, and **exponentially decay** occurs when $k < 0$.

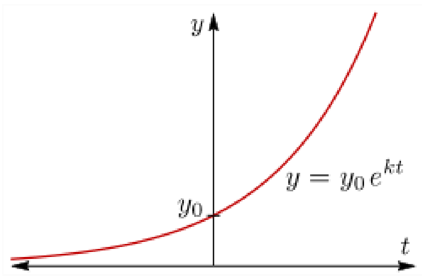
Both exponential growth and exponential decay are quite common in science, we will discuss several applications of these in sections.

Exponential growth

Exponential growth describes a hypothetical model for population growth in which space and resources are available in unlimited supply. As a result, the population growth rate increases with each new generation. Under this model, a population rapidly grows quite large, and continues to grow indefinitely.

THE EXPONENTIAL GROWTH EQUATION:

The exponential growth equation is the differential equation $\frac{dy}{dx} = ky, (k > 0)$. Its solutions are exponential functions of the form $y = Ce^{kt}$, where $y_0 = y(0) = C$ is the initial value of y. Fig. 1 shows the graph of a typical exponential function, assuming $y_0 > 0$ and k>0. Because of the factor of e^t, an exponential function increases quite quickly as 't' increases, as illustrated in Fig. 2.



▲ **Figure 1:** Exponential growth.

t	e^t
1	2.72
2	7.39
5	148.41
10	22,026.47
20	485,165,195.41

▲ **Figure 2:** The exponential function e^t grows very quickly as t increases.

THE GROWTH CONSTANT:

The constant 'k' in the equations $\frac{dy}{dx} = ky$ and $y = Ce^{kt}$ is called the growth constant or exponential growth rate. It controls how rapidly the exponential function grows – higher values of 'k' correspond to faster growth, while lower values of 'k' correspond to more gradual growth.

DOUBLING TIME:

Doubling time is the amount of time it takes for a given quantity to double in size or value at a constant growth rate.

PROBLEMS:

1. Today there are 1000 birds on an island. They breed with a constant continuous growth rate of 10% per year. How many birds will be on the island after seven years?

Solution:

Let 'x' be the number of birds on the island after 't' years. Given x(0)=1000.

With continuous growth rate

k = 10% = 10/100 = 0.1, we have

$\frac{dx}{dt} = 0.1x$ and hence solving the differential equation, $x(t) = Ce^{0.1t}$, for some positive constant 'C'. Since x(0)=1000, we have C = 1000, so

$$x(t) = 1000e^{0.1t}$$

After seven years, the number of birds on the island is $x(7) = 1000e^{0.1x7} = 2013.75 \approx 2014$

2. A bacteria culture is known to grow at a rate proportional to the amount present. After one hour, 1000 strands of the bacteria are observed in the culture; and after four hours, 3000 strands. Find (a) an expression for the approximate number of strands of the

bacteria present in the culture at any time 't' and (b) the approximate number of strands of the bacteria originally in the culture.

Solution:

Let N(t) denote the number of bacteria strands in the culture at time 't'.

$\Rightarrow \dfrac{dN(t)}{dt} - kN = 0,$ which is both linear and separable. Its solution is $N(t) = Ce^{kt}$

At t=1, N = 1000; hence, $1000 = Ce^{k(1)}$ --------→(1)

At t=4, N = 3000; hence, $3000 = Ce^{k(4)}$ --------→(2)

Solving eq. (1) & (2) for 'k' and 'C' , we find

Divide eq.(2) by (1) , we get

$$\frac{3000}{1000} = \frac{Ce^{4k}}{Ce^{k}} \Rightarrow 3 = e^{3k}$$

Taking ln on both sides, we get

$$\ln e^{3t} = \ln 3$$
$$3k = \ln 3$$
$$\Rightarrow k = \frac{\ln 3}{3} = 0.3662$$

Substitute the value of 'k' in eq.(1), we get

$$1000 = Ce^{(0.3662)}$$
$$\Rightarrow C = 1000 \times e^{-0.3662} = 693.36 \approx 693$$

Subsitute the values of 'C' and 'k' in $N(t) = Ce^{kt}$

$$N(t) = 693e^{0.3662t} \quad \text{-------→(3)}$$

as an expression for the amount of the bacteria present at any time t.

We require N at t= 0. Substituting t=0 into (3) we obtain

$$N(0) = 693e^{0.3662(0)} = 693$$

3. A person places \$5000 in an account that accrues interest compounded continuously. Assuming no additional deposits or withdrawals, how much will be in the account after seven years if the interest rate is a constant 8.5 % for the first four years and a constant 9.25% for the last three years?

Solution:

Let N(t) denote the balance in the account at any time ‘t’. Initially, N(0) = 5000. For the first four years, $k = 8.5\% = \frac{8.5}{100} = 0.085$ and the differe. equation is,

$$\frac{dN}{dt} = kN \Rightarrow \frac{dN}{dt} - 0.085N = 0$$

Its solution is $N(t) = Ce^{kt} \Rightarrow N(t) = Ce^{0.085t}, (0 \le t \le 4)$ -------------→(1)

At t=0, N(0)=5000, which when substituted into (1) yields

$$5000 = Ce^{0.085(0)} \Rightarrow C = 5000$$

And eq.(1) becomes, $N(t) = 5000e^{0.085t}, (0 \le t \le 4)$ -------------→(2)

Substitute t = 4 into (2), we find the balance after four years to be

$$N(4) = 5000e^{0.085(4)} = 5000(1.40495) = \$7024.74$$

This amount also represents the beginning balance for the last three-year period.

Over the last three years, the interest rate is 9.25 % and the equ. becomes,

$$\frac{dN}{dt} = kN \Rightarrow \frac{dN}{dt} - 0.0925N = 0, (4 \le t \le 7)$$

Its solution is $N(t) = Ce^{kt} \Rightarrow N(t) = Ce^{0.0925t}, (4 \le t \le 7)$ --------------→(3)

At t =4, $N(4) = \$7024.74$, which when substituted into (3) yields

$$7024.74 = Ce^{0.0925(4)} = C(1.447735) \Rightarrow C = 4852.23$$

And equ. (3) becomes

$$N(t) = 4852.23e^{0.0925t}, (4 \le t \le 7)$$ --------------→(4)

Substituting t = 7 into equ.(4), we find the balance after seven years to be

$$N(7) = 4852.23e^{0.0925(7)} = 4852.23(1.910758) = \$9271.44$$

4. What constant interest rate is required if an initial deposit placed into an account that accrues interest compounded continuously is to double its value in six years?

Solution:

The balance N(t) in the account at any time 't' is governed by $\frac{dN}{dt} = kN$.

Which has its solution $N(t) = Ce^{kt}$ ----------------------→(1)

We are not given an amount for the initial deposit, so we denote it as N_0.

At t =0, $N(0)=N_0$, which when substituted into eq.(1) gives $N_0 = Ce^{k(0)} \Rightarrow C = N_0$

And eq.(1), becomes $N(t) = N_0 e^{kt}$ --------------------------→(2)

We seek the value of 'k' for which $N=2N_0$ when t = 6. Substituting these values into equ.(2) and solving for 'k', we find

$$2N_0 = N_0 e^{k(6)}$$
$$\Rightarrow e^{6k} = 2$$

Taking ln on both sides, we get

6k=ln 2 $\Rightarrow k = \frac{\ln 2}{6} = 0.1155$

An interest rate of 11.55 % is required.

Practice Problems:

1. A person places \$20,000 in a savings account which pays 5 % interest per annum, compounded continuously. Find (a) the amount in the account after three years, and (b)the time required for the account to double in value, presuming no withdrawals and no additional deposits.
2. The population of a certain country is known to increase at a rate proportional to the number of people presently living in the country. If after two years the population has doubled, and after three years the population is 20,000, estimate the number of people initially living in the country.

Discussion Questions

Try to answer the following questions

1. Define exponential growth.
2. What is doubling time?
3. Give a real time example of exponential growth.
4. Let $y=P(t)$ denote the number of fruit flies in the kitchen, 't' days since you first observed them. It is known that this species of flies exhibits exponential growth with growth constant $k=.3$. Suppose that the initial number of fruit flies is 6.

(a) Find $P(t)$

(b) Estimate the number of fruit flies after 7 days.

5. A colony of fruit flies grows at a rate proportional to its size. At time $t=0$, approximately 20 fruit flies are present. In 5 days there are 400 fruit flies. Determine a function that expresses the size of the colony as a function of time, measured in days.
6. Approximately 10,000 bacteria are placed in a culture. Let $P(t)$ be the number of bacteria present in the culture after 't' hours, and suppose that $P(t)$ satisfies the differential equation $P'(t)=0.55P(t)$
   (a) What is $P(0)$?
   (b) Find the formula for $P(t)$.
   (c) How many bacteria are there after 5 hours?
   (d) What is the growth constant?
   (e) What is the size of the bacteria culture when it is growing at a rate of 34,000 bacteria per hour?